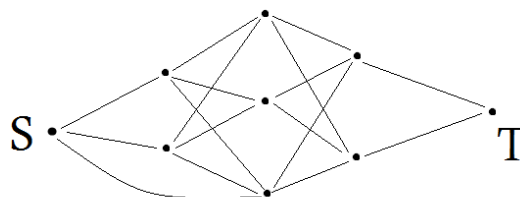
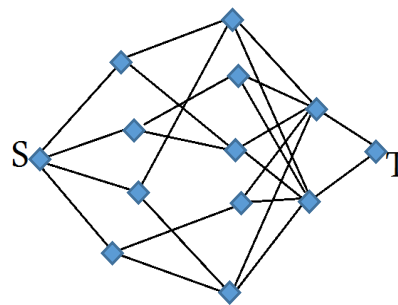


1. A bistro menu has 4 starters, 11 mains, and 5 desserts.
 - a. Alfred wants three courses. How many different possible meals could he have?
 - b. Boris wants two courses, but he is happy with either starter/main or main/dessert. How many possible meals could Boris have?
 - c. Charles isn't fussy. He asks the waiter to choose for him. The waiter asks "certainly sir – how many courses would you like?", to which Luke replies "up to you – as long as I get some food, and don't have the same course twice, I'll be satisfied!". How many possible meals could Luke be served?
2. Three academic prizes (in English, Geography, and Chemistry) are to be awarded within a year of 120 boys. The respective Heads of Department agree that no boy will receive more than one prize. In how many ways could the prizes be awarded?
3. How many anagrams of DUCKLING are there? How about SANDWICH?
4. Eight sprinters, including Janice, ran a 100 m race.
 - a. In how many different ways could the bronze, silver, and gold medals be awarded?
 - b. In how many different orders could they finish?
 - c. Suppose that Janice placed 4th. In how many ways could the three medals be awarded?
 - d. Suppose that Janice won a medal. In how many ways could the three medals be awarded?
5. How many different routes from S to T are there in this diagram? (You can only move from left to right).
6. In a far away land, people whose birthdays are in odd months can only marry people whose birthdays are in even months. Twelve people in a group have their birthdays such that each month of the year is covered by someone's birthday. In how many different ways can they be paired off?
7. How many routes from S to T? (Again, you can only move from left to right)
8. How many arrangements of the letters in the word DUCKLING:
 - a. start with a vowel?
 - b. end with a consonant?
 - c. start with a vowel and end with a consonant?
 - d. have all of the vowels together?
9. A ternary string consists of a list of 0's, 1's and 2's, which must not start with a 0. For example, 22012 is a ternary string of length 5 (but 01102 is not).
 - a. How many ternary strings of length five are there?
 - b. How many ternary strings of length at most five are there?
 - c. Can you find a formula for the number of ternary strings of length at most n ?



10. A falling number is an integer whose decimal representation has the property that each digit except the units digit is larger than the one to its right. For example, 96521 is a falling number but 89642 is not.
- a) How many single digit falling numbers are there?
 - b) How many two digit falling numbers are there?
 - c) How many three digit falling numbers are there?
 - d) Can you generalise this for an n digit number?